

Clinical Readmission Pipeline - Executive Summary (v2)

Steps 1-3: EDA -> Cleaning -> Feature Engineering | Target: Readmission_30d | 2026-09-02

Step 1 - Data Collection & Exploration

Records loaded:

1,155,000 patients | 41 raw features

Target balance:

Not Readmitted: 84.98% (910,243) **Readmitted: 15.02% (160,918) - imbalanced**

Key issues found:

Duplicate case-mismatched columns (Age/age, Gender/gender, Drug_Name/drug_name); 100% null columns (Extra_Col_1/2); invalid values (Age up to 999, negative weights); inconsistent Gender entries; 50+ Drug_Name typo variants.

Outputs generated:

step1_clinical_distributions.png, step1_correlation_matrix.png

Step 2 - Data Cleaning & Preprocessing (re-executed)

- Column reduction: dropped 11 columns (nulls, duplicates, redundant) -> 41 to 30 columns
- Row validation: removed 37,988 rows with invalid Age/Weight
- Missing target rows: dropped 81,110 rows without a Readmission_30d label
- Gender fix: standardized to 4 clean categories - Male 276,523 | Female 276,586 | Other 117,968 | Unknown 445,935 (previously 10+ inconsistent raw tokens)
- Drug_Name fix: typos normalized (e.g., amoxicilin/amoxycillin -> amoxicillin) and grouped into 22 therapeutic classes (Antibiotic, Statin, ACE Inhibitor, Insulin, NSAID, etc.); raw Drug_Name dropped in favor of Drug_Class
- Imputation: numeric columns filled with median, categorical with mode
- Duplicates removed: 63,982 duplicate rows dropped
- Outlier capping (IQR): Age, BMI, Dosage, Hemoglobin, Creatinine, ALT/AST enzymes
- Encoding: label-encoded Gender; one-hot encoded Drug_Class and Route (was raw Drug_Name)

Final output: data_cleaned_v2.csv -> 971,920 rows x 183 columns (was 553)

Watch-outs resolved: Gender inconsistency and Drug_Name cardinality issues from the prior report have both been fixed in this re-execution.

Step 3 - Feature Engineering (re-executed)

kidney_stage	eGFR-based staging (Normal/Mild/Moderate/Severe)
liver_risk	Flag if ALT or AST > 40
polypharmacy	Flag if 5+ concurrent medications
bmi_category	Underweight/Normal/Overweight/Obese
age_group	Pediatric through Elderly bands
elderly_high_dose	Age > 65 and above-median dosage
de_ritis_ratio	AST/ALT ratio (liver disease indicator)

Final output: data_engineered_v2.csv -> 971,920 rows x 190 columns (was 560)

Pipeline Status

Step	Status	Runtime Notes
1. EDA	Complete	No errors
2. Cleaning (v2)	Complete	Gender + Drug_Name fixes applied; minor deprecation warnings only
3. Feature Eng. (v2)	Complete	Performance warnings (fragmentation) only, no data loss

Next Step: Step 4 - Model Training & Evaluation on data_engineered_v2.csv